

BRYAN WEI XUAN CHEONG

+44 7825 787529 | bryancheong.dev | wei.cheong24@imperial.ac.uk | bryancwx | bryan-cheong

EDUCATION

Imperial College London – MSc in Business Analytics (Predicted Distinction) Aug 2024 - Aug 2025
Modules: Generative AI & LLMs, Statistics & Econometrics, Optimisation & Decision Models, Financial Analytics, Text Analytics, Digital Marketing Analytics, Network Analytics, Energy Analytics, Logistics & Supply Chain Analytics, Data Visualisation

University College London – BSc in Computer Science (2:1, Grade 68%) Sep 2021 - Jul 2024
Modules: Machine Learning, Reinforcement Learning, Mathematics & Statistics, Algorithms, OOP, Software Engineering, Security

SKILLS

Programming	Python, SQL, R, JavaScript (React, SvelteKit), HTML, TailwindCSS (Proficient); Java, Haskell, C (Basic)
ML Libraries	pandas, OpenAI, pydantic, scikit-learn, XGBoost, PyTorch, FastText, SHAP, statsmodels, PyMuPDF, NetworkX
Technical Tools	Git, AWS, FastAPI, Docker, Spark, PostgreSQL, Power BI, VWO, Google Analytics, Google Docs API, Selenium
Finance	Mathematical Modelling, Forecasting, Portfolio Optimisation, Monte Carlo Simulation, CVar, Hedging
Languages	English (Native), Cantonese (Fluent), Mandarin (Intermediate), Malay (Intermediate)

WORK EXPERIENCE

Tecno International (Capstone Project, Imperial College London) London, UK
AI Agent Engineer – Multimodal ESG Copilot for SMEs Jun 2025 - Present

- Deployed an AI assistant that automates ESG assessments from SMEs’ PDF/HTML files through GPT-4.1 agents; flagged underperforming ESG criteria and explained rationale behind each score
- Prototyped a Research Agent using RAG on internal corpora and web search to complete a 10-point checklist on business impact
- Generated three ESG reports via Google Docs API, cutting analysis time by 80% and freeing consultants for higher-order work
- Optimised token usage and implemented response caching for FAQs, reducing cost by ~40% without loss of output quality

Anavio Capital Hedge Fund London, UK
Equity Analytics Intern – Long/Short Strategy Jul 2023 - Sep 2023

- Built a Python dashboard to uncover patterns in firm-level efficiency across 15+ metrics, identifying outliers for investment decisions
- Scaled coverage from 10 to 200 tickers and automated 80% of manual analysis, accelerating idea generation for the portfolio manager
- Enhanced pipeline robustness with 100+ Pytest tests and six error-handling decorators, minimising production risk and downtime
- Translated business rules into flowcharts to clarify logic and align tool design with non-technical stakeholders, halving delivery time

GSE-M Technology Sdn. Bhd Shah Alam, Malaysia
Enterprise Resource Planning (ERP) Assistant May 2023 - Jul 2023

- Partnered with ERP users across five departments to gather 30+ requirements, identify operational challenges, and find critical bugs
- Tailored ERP functionalities with consultants, improving process efficiency by 25% and aligning system with departmental needs
- Co-led daily conference room pilots, training 90 employees to ensure smooth adoption and usage of the new ERP system

British Red Cross London, UK
Data Analyst Sep 2022 - Apr 2023

- Developed 20+ Power BI dashboards to visualise behavioural trends in complaints across the UK for operational and senior teams
- Defined resolution KPIs and identified delay drivers, reducing resolution time by 50% and improving service outcomes for customers
- Built an ETL pipeline from four sources and used DAX for geographical drill-downs, enabling scalable and customised reporting
- Secured future development by presenting a streamlined issue resolution system along with a deployment manual to the Head of Data

Subplace (B2C Subscription E-commerce Platform) Kuala Lumpur, Malaysia
Software Reliability Intern Jun 2022 - Aug 2022

- Cut maintenance time by 60% via modular and systematic test case templates, improving stability for a SQL-backed platform
- Resolved 200+ issues and introduced agile, test-driven development practices to enhance reliability and support continuous delivery

PROJECTS

RecSys Purchase Prediction. Built a two-stage pipeline on 33M clicks and 1.1M purchases to predict purchasing sessions with XGBoost and forecast items bought via heuristic ranking. Engineered 21 features with temporal encodings and rarity filters. Tuned thresholds to optimise a Jaccard-based score (10,491) and used SHAP to interpret key drivers like session duration and item popularity

Multi-Channel Attribution. Analysed 3.3M contacts and 240K orders to compare catalogue vs. email campaign effectiveness with PostgreSQL and Python. Computed attribution models (first-, last-, linear-click), revealing catalogue’s six times higher conversion. Segmented customers into 125 RFM cells, estimated CLV, and recommended targeting strategies based on breakeven cost and ROI

Portfolio A/B Testing and Analytics. Ran A/B tests with VWO to measure impact of clickbait vs professional phrasing on my portfolio website’s “Download CV” CTA; the professional version saw a 70.3% higher CTR (p = 0.043, statistically significant). Drove traffic via social media, Selenium bots, and a targeted Google Ads campaign; tracked conversions and bounce rate using event-based analytics

Beat The Bookies. Engineered a Premier League match predictor using 18 features across five seasons (older data degraded performance), achieving 51% validation and ~40% test accuracy with CatBoost. Feature-engineered Pi-rating variants with tuned hyperparameters and selected key variables using SHAP. Explored pre-game sentiment as a predictive signal using LLMs and Nitter API

RIE Covariance Cleaning. Developed a spectral cleaning algorithm using Rotationally Invariant Estimators, shrinking noise in high-dimensional lagged covariance matrices. Achieved 100% noise reduction in AR(0) and ~20% signal-to-noise improvement in AR(1), with more gains as dimensionality increased. Applicable to portfolio risk modelling, time series forecasting, and signal processing

Text Analytics on Song Lyrics. Modelled 40,000+ lyrics through LASSO regression with n-grams and FastText embeddings to explore song popularity drivers. Discovered eight genre-level trends in explicitness and thematic content. Leveraged PCA-filtering and cosine similarity on word vectors to quantify semantic shifts between artists’ first albums and subsequent works

Logistics and Supply Chain Optimisation Game. Simulated a multi-region supply chain implementing hybrid demand forecasting (SARIMA, Croston’s method) and cost-benefit analysis to guide factory and warehouse expansion. Applied adjusted reorder points and Silver-Meal heuristics to manage seasonal inventory, securing 2nd place in Imperial’s cohort

Energy Demand Forecasting Model. Developed and tuned XGBoost model to forecast UK electricity demand, engineering features like Heating Degree Days, holiday dummies, and temporal lags. Achieved 96.3% R²; generalises well with low residual autocorrelation

Gas Plant Risk & Hedging. Simulated 10,000 copula-based gas and electricity prices to assess plant risk and profitability. Designed a hedge (long electricity, short gas) that cut 95% Expected Shortfall by 29%, factoring in outages and cross-commodity correlations

Gas Infrastructure Valuation with Real Options. Valued a two-stage gas project using NPV and real options. Identified a break-even gas price of \$0.136/m³ and option values of \$1.25M (start-of-year) and \$1.15M (mid-year), highlighting the effect of timing on valuation. Deferring Stage 2 added \$1.25M in value, demonstrating the financial benefits of flexibility under uncertainty

TetrisAI. Implemented a Tetris heuristics evaluation AI, scored 28,000+ points across 400 blocks, 80th percentile in UCL’s cohort

AWARDS AND CERTIFICATIONS

AmplifyME – Mergers and Acquisition Simulation with UBS	Feb 2025
Boston Consulting Group – Introduction to Strategy Consulting (Forage)	Oct 2024
Goldman Sachs – Software Engineering Experience Program (Forage)	Jul 2024
Harvard University – CS50x Puzzle Day: Perfect Score (Top 13% Worldwide, Teams of Six)	Apr 2020
Land Rover 4x4 In Schools Challenge – National Champions	Nov 2019
Global Space Design Challenge – Innovation Award	Jul 2019